

Akash Gogate

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EDUCATION

University of Wisconsin at Madison

Graduating: May 2028

Bachelor of Science in Computer Science • Dean's List Fall 2024, Spring 2025, Fall 2025 • 3.9 GPA

Coursework: Artificial Intelligence, Deep Learning, Advanced Algorithms, Machine Organization and Programming, Cryptography, Data Structures, Computer Science Capstone, AP Computer Science A, AP Statistics, Game Development, Web Development

TECHNOLOGIES

Languages: Python, R, Java, JavaScript, Rust, C++, SQL, Linux/Bash

ML & AI: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, Sentence-Transformers, Hugging Face, Transformer Architectures, Neural Networks, Deep Learning, NLP, Natural Language Processing, Generative AI, Random Forest, KNN, Feature Engineering, Hyperparameter Tuning, Model Fine-Tuning, Inference Optimization, Model Deployment

Agentic & LLM: LLM Agents, LLM APIs, GPT-4o, RAG, Semantic Similarity Search, Vector Databases (Supabase, Pinecone, Qdrant), FastAPI, MLflow, Weights & Biases, A/B Testing, Benchmarking, Evaluation, Experiment Tracking, Model Monitoring, Model Serving, Inference Serving

GPU & MLOps: CUDA, GPU-Accelerated Computing, Docker, Kubernetes, Kafka, MongoDB, CI/CD, GitHub Actions, Git, Linux, pytest, Scalable Production Systems

EXPERIENCE

The Kendziorz Lab at the University of Wisconsin-Madison

September 2025 - Present

Machine Learning Research Intern

- Built end-to-end ML pipeline for scRNA-seq + spatial transcriptomics using PyTorch/TensorFlow deep learning; agentic framework selected and evaluated models across scVI/scANVI/scGen/AmortizedLDA for scalable production genomics, **reducing GPU computation 300%**; tracked via Weights & Biases
- Engineered TransferAgent automating natural language processing (NLP) literature synthesis and cross-paper validity scoring; cut review time by **~60%**; collaborated with clinicians on clinical genomics impact
- Validated model performance and pipeline reproducibility across interdisciplinary research team; findings contributed to ongoing clinical genomics publication and drug discovery workflows

Leidos

May 2025 - Present

Software Engineer Intern

- Built fault-tolerant Kafka + MongoDB event pipeline with **at-least-once delivery guarantees**; optimized scheduling via memoization + DP for mission-critical ops
- Deployed Docker-containerized ML-adjacent test infrastructure on Kubernetes via **GitHub Actions CI/CD** with full component coverage

Inspirit AI Research Internship

Sep 2022 - Mar 2023

Machine Learning Research Intern

- Built **skin cancer detection pipeline** (OpenCV + scikit-learn/TensorFlow, NumPy/Pandas) achieving **93% classification accuracy**; published as **peer-reviewed paper** through Inspirit AI
- Trained Random Forest on miRNA sequences achieving **95% predictive accuracy ($p < 0.05$)**; validated via stratified k-fold cross-validation

TECHNICAL PROJECTS

Self-Improving LLM Agent (<https://github.com/AkashGogate/SelfImprovingLLMAgent>)

Jan 2026 - May 2026

- Architected failure-driven self-improvement loop across **5,000+ task benchmark**, outperforming ReAct by **15pp** via RAG-enhanced retrieval and Transformer neural network reasoning
- Optimized inference via fine-tuning, task horizon tuning + parameter search; built provider-agnostic serving abstraction supporting end-to-end ML pipeline scaled to horizon 20
- Engineered LLM-as-judge harness scoring 7 dimensions; implemented novel **repeated failure rate** metric to quantify self-improvement efficacy

Lotus Health (<https://github.com/AkashGogate/lotus-health>)

March 2026

- Built probabilistic disease cost engine on **1,080-node ICD-10 graph** across **45M records**; RAG pipeline with semantic similarity search over graph as vector database
- Designed hybrid architecture with model deployment pipeline: Groq Llama NLP inference layer + deterministic Python simulation engine; ensures **hallucination-free, ACID-compliant** outputs

miRcore Cancer Detection Research

Jul 2021

- Trained and validated Random Forest classifier in R on miRNA sequences achieving **95% predictive accuracy ($p < 0.05$)**; validated via stratified k-fold cross-validation for cancer cell likelihood prediction

LEADERSHIP

Princeton Racket Club

May 2024 - Aug 2024

Tennis Coach and Tournament Director

- Directed tournament logistics; coached athletes across skill levels in cross-functional environment; managed end-to-end client communications

Tennis Racket Stringing Services

Jan 2019 - Present

- Founded and scaled to **50+ loyal clients**; manage all operations, inventory, and client consultations